

## Lesson 07.2: Posterior-based Decisions

**Instructor:** Dr. GP Saggese, [gsaggese@umd.edu](mailto:gsaggese@umd.edu)

**References:**

- AIMA (Artificial Intelligence: a Modern Approach)
  - Chap 15: Probabilistic programming
- Martin, Bayesian Analysis with Python, 2018 (2e)



- *Posterior-Based Decisions*
  - Chemical Shift: Example
  - Posterior Predictive Checks
  - Groups Comparison

# Posterior-Based Decisions

---

- Sometimes describing the posterior is not enough
  - **Make decisions based on inference**
- E.g., is the coin fair ( $\theta = 0.5$ ) or biased?
  - $\mathbb{E}[\hat{\theta}] = 0.324$  suggests bias
  - Can't rule out unbiased since:
    - \*  $HPI = [0.03, 0.65]$
    - \*  $0.5 \in HPI$
  - For **sharper decisions**:
    - \* Collect more data to reduce posterior spread
    - \* Define a more informative prior

# Savage-Dickey Density Ratio

- **Savage-Dickey ratio** tests *point null-hypotheses* in Bayesian inference
- **Idea**: compare prior and posterior densities at a single point  $\theta_0$

$$BF_{01} = \frac{p(\theta_0|H_1)}{p(\theta_0|\mathcal{D}, H_1)}$$

where:

- $p(\theta_0|H_1)$  is *prior* density  $\theta$  under alternative hypothesis  $H_1$ , evaluated at  $\theta_0$
- $p(\theta_0|\mathcal{D}, H_1)$  is *posterior* density  $\theta$  under  $H_1$  evaluated at  $\theta_0$

- **Intuition**: show how data changes belief about  $\theta_0$

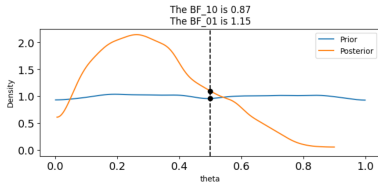
- If posterior density at  $\theta_0$  is much smaller than prior density, strong evidence against null hypothesis  $H_0$

| Bayes Factor (BF) | Interpretation       |
|-------------------|----------------------|
| 1 - 3             | Not enough evidence  |
| 3 - 10            | Substantial evidence |
| 10 - 100          | Strong evidence      |
| > 100             | Decisive evidence    |

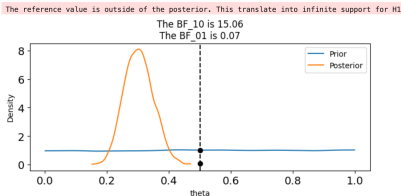
- **Cons**:
  - It's a point statistics (not considering the entire posterior)

# Savage-Dickey Density Ratio: Example

```
[29]: az.plot_bf(idata1, var_name="theta", prior=np.random.uniform(0, 1, 10000), ref_val=0.5);
```



```
[30]: az.plot_bf(idata2, var_name="theta", prior=np.random.uniform(0, 1, 10000), ref_val=0.5);
```



- $H_0$ : "coin is fair"
  - The prior is 0.87
  - The posterior is 1.15
  - $BF_{10} = 0.76 \rightarrow$  **no evidence** to reject the null hypothesis
- In the other case
  - $BF_{10} = 15.06 \rightarrow$  **strong evidence** to reject the null hypothesis

# Region of Practical Equivalence

---

- **Region of Practical Equivalence** (ROPE) = interval where values are “equivalent”
- **Example**
  - $H_0$ : “coin is fair” impractical if  $\theta = 0.5$
  - ROPE:  $\theta \in [0.45, 0.55]$  equivalent to 0.5
- **Hypothesis testing with ROPE and HPI**
  - Compare ROPE with HPI (Highest-Posterior Interval)
    - \* HPI within ROPE  $\rightarrow$  no effect, reject  $H_1$
    - \* HPI outside ROPE  $\rightarrow$  effect present, reject  $H_0$
    - \* HPI overlaps ROPE  $\rightarrow$  inconclusive
- **Decide ROPE before analysis based on domain knowledge**
  - Picking it after analysis is like choosing p-value threshold after seeing p-value

# Loss Function: Motivation

---

- For many problems:
  - **Make decisions based on inference**
  - **Decision cost is asymmetric**
    - \* Cost of a bad decision  $>$  or  $<$  benefit of a good decision
    - \* Vaccines may cause overreaction, but benefits outweigh risks
- Measure for **best decision**:
  - Benefits of a correct decision
  - Cost of a mistake
  - Trade-off between benefits and costs using a loss function
  - Use loss function for decisions

# Loss Function

- Aka “cost function”
  - The inverse is known as “objective”, “fitness”, “utility function”
- **Loss function** quantifies “*how bad is an estimation mistake?*”
  - Larger loss indicates worse estimation
  - Loss is difference between:
    - \* The true value  $\theta$ ; and
    - \* The estimated value  $\hat{\theta}$

| Loss           | Expression                    | Point estimate      |
|----------------|-------------------------------|---------------------|
| Quadratic loss | $(\theta - \hat{\theta})^2$   | Mean of posterior   |
| Absolute loss  | $ \theta - \hat{\theta} $     | Median of posterior |
| 1-0 loss       | $I(\theta \neq \hat{\theta})$ | Mode of posterior   |

- Algorithm to make decisions in Bayesian statistics using loss function
  - Goal: pick a single value  $\hat{\theta}$
  - True value  $\theta$  is unknown
  - Estimate  $\theta$  using posterior distribution
  - Find  $\hat{\theta}$  minimizing expected loss function



# Tutorial

---

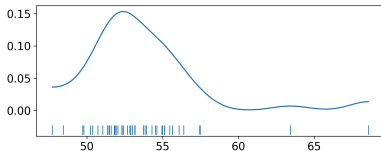
- [Bayesian Coin](#)

- Posterior-Based Decisions
  - *Chemical Shift: Example*
  - Posterior Predictive Checks
  - Groups Comparison

# Chemical Shift

---

- **Nuclear magnetic resonance** (NMR)
  - Used to study molecules of living things
  - Measures observable quantities related to unobservable molecular properties (e.g., chemical shift)
- **Chemical shift** reveals local magnetic environment of atomic nuclei
  - Identifies molecular structure and functional groups
- **Example of chemical shift**
  - Data looks Gaussian with a couple of outliers



# Use of Gaussians in Statistics

---

- Aka “normal” distribution
- **Gaussians** are:
  - easy to work with
  - abundant in nature
- **Pros:**
  - Average of large sample size tends to be Gaussian (Central Limit Theorem)
  - Many phenomena approximated using Gaussians (since they are average of effects)
  - Conjugate prior of Gaussian is Gaussian
- **Cons:**
  - Not robust to outliers
  - Important to relax assumption of Gaussianity

# Chemical Shift: Example

- **Assume Gaussian** approximates data from example chemical shift
- **Likelihood** from normal distribution:

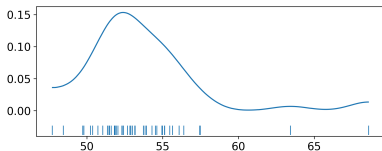
$$Y \sim N(\mu, \sigma)$$

- Weakly informative **priors** for mean and sigma of  $Y$ 
  - **Mean** from uniform distribution

$$\mu \sim U(l = 40, h = 70)$$

- \* Larger than data range, e.g.,  $[40, 70]$
- **Std dev** from half-normal distribution

$$\sigma \sim \text{HalfNormal}(0, \sigma_\sigma = 10)$$



# Chemical Shift: PyMC

```
[87]: with pm.Model() as model_g:
    # The mean is Uniform in [40, 70] (which is larger than the data).
    mu = pm.Uniform("mu", lower=40, upper=70)
    # The std dev is half normal with a large value (which is a large value based on the data).
    sigma = pm.HalfNormal("sigma", sigma=10)
    # The model is N(mu, sigma).
    y = pm.Normal("y", mu=mu, sigma=sigma, observed=data)
    # Sample.
    idata_g = pm.sample(1000)
```

```
Auto-assigning NUTS sampler...
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [mu, sigma]
```

Sampling 4 chains, 0 divergences  100% 0:00:00 / 0:00:00

Sampling 4 chains for 1\_000 tune and 1\_000 draw iterations (4\_000 + 4\_000 draws total) took 1 seconds.

- The PyMC code is **one-to-one with the model**:

$$\begin{cases} \mu \sim U(l = 40, h = 70) \\ \sigma \sim \text{HalfNormal}(0, \sigma_\sigma = 10) \\ Y \sim N(\mu, \sigma) \end{cases}$$

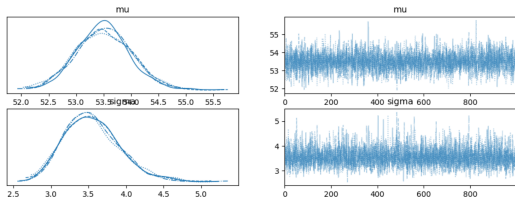
- Get 1000 samples from the posterior

# Chemical Shift: PyMC

- Compute 4 traces for 2 variables  $\mu$ ,  $\sigma$ 
  - Results are well-formed
  - $\mu \in [52.55, 54.45]$
  - $\sigma \in [2.86, 4.23]$
- Is the model good?

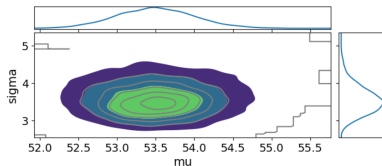
```
[92]: # There are 4 traces for 2 variables.
```

```
az.plot_trace(idata_g);
```



```
[93]: # The posterior distribution of the params is bi-dimensional, since it has mu and sigma.
```

```
az.plot_pair(idata_g, kind='kde', marginals=True);
```



```
[94]: # Report a summary of the inference.
```

```
az.summary(idata_g, kind='stats').round(2)
```

```
[94]:
```

|       | mean  | sd   | hdi_3% | hdi_97% |
|-------|-------|------|--------|---------|
| mu    | 53.50 | 0.51 | 52.55  | 54.46   |
| sigma | 3.55  | 0.38 | 2.86   | 4.23    |

- Posterior-Based Decisions
  - Chemical Shift: Example
  - *Posterior Predictive Checks*
  - Groups Comparison



# Samples from Posterior Distribution

---

- Given **posterior distribution**  $\Pr(\theta|y)$ , generate predictions  $\tilde{y}$  based on data  $y$  and estimated parameters  $\hat{\theta}$ :

$$\Pr(\tilde{y}|y) = \int_{\hat{\theta}} \Pr(\tilde{y}|\theta) \Pr(\theta|y) d\theta = \int \text{model} \times \text{posterior}$$

- This is called the “*posterior predictive distribution*” as it predicts *future data* using the *posterior distribution*
- **Conceptually**
  - Sample a value of  $\theta$  from the posterior  $\Pr(\theta|y)$
  - Feed the value of  $\theta$  to the likelihood  $\Pr(\tilde{y}|\theta)$
  - Obtain  $\tilde{y}$
- This process has **two sources of uncertainty**:
  - Parameter uncertainty
    - \* Captured by the posterior  $\Pr(\theta|y)$
  - Sampling uncertainty
    - \* Captured by the likelihood  $\Pr(\tilde{y}|\theta)$

# Posterior Predictive Check (PPC)

---

- **PPC approach:**
  - Generate predictions  $\tilde{y}$  with observed data  $y$  from posterior distribution
  - Check consistency between predictions and observed data
- **Intuition:** can the model reproduce observed data?
  - Always check!
- **Differences** arise from:
  - Mistakes
  - Model limitations
    - \* E.g., works for average behavior but fails for rare values
  - Data limitations

# Bayesian Workflow Using PPC

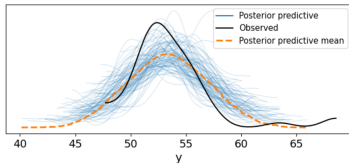
---

1. Given a process, **true distribution** is unknown / unknowable
2. **Sample the process**
  - Get finite sample  $y$
  - E.g., experiment, survey, simulation
3. **Inference**
  - Build probabilistic model using prior  $\Pr(\theta)$  and likelihood  $\Pr(y|\theta)$  to get posterior  $\Pr(\theta|y)$
  - Posterior is distribution of model parameters  $\theta$  given data
4. **Predictive distribution**
  - Compute predictions from posterior (posterior predictive distribution)
  - Posterior predictive is distribution of predicted samples averaged over posterior
5. **Validation**
  - Validate model by comparing original vs predicted samples

# Chemical Shift Example: PPC

```
[95]: # Compute 100 posterior predictive samples.  
y_pred_g = pm.sample_posterior_predictive(idata_g, model=model_g)  
  
Sampling: [y]  
  
Sampling ... 100% 0:00:00 / 0:00:00
```

```
[96]: # Black: KDE of the data (observed)  
# Blue: KDEs of the posterior predictive samples  
# Orange: KDE of the posterior predictive mean  
az.plot_ppc(y_pred_g, mean=True, num_pp_samples=100);
```



- **Apply the Bayesian workflow**
  - Sample posterior
  - Apply model
  - Get predictive posterior distribution (dashed orange)
  - Compare to data (black)
- **Is the PPC model good?**
  - **No!**
  - Posterior mean is right of data
  - Posterior std dev is larger
  - You can't generate the data from the model!

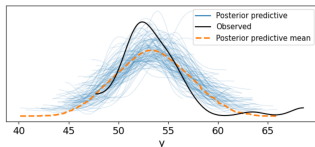
# Chemical Shift: Model Critique

## ■ Problem

- Two data points on distribution tails
- Normal distribution:
  - \* is “surprised” by these points
  - \* “reacts” by adjusting mean towards them and increasing the standard deviation

```
[95]: # Compute 100 posterior predictive samples.  
y_pred_g = pm.sample_posterior_predictive(idata_g, model=model_g)  
  
Sampling: [y]  
  
Sampling ... 100% 0:00:00 / 0:00:00
```

```
•[96]: # Black: KDE of the data (observed)  
# Blue: KDEs of the posterior predictive samples  
# Orange: KDE of the posterior predictive mean  
az.plot_ppc(y_pred_g, mean=True, num_pp_samples=100);
```

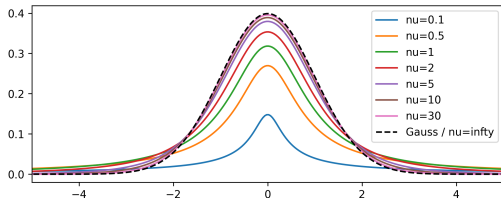


## ■ Possible solutions

1. Declare and discard outliers
  - E.g., equipment malfunction (requires evidence)
2. Change the model
3. **Bayesian philosophy**
  - Encode assumptions into the model (e.g., priors, likelihoods)
  - Avoid ad-hoc heuristics (e.g., outlier removal rules)

# Student's t-distribution: Recap

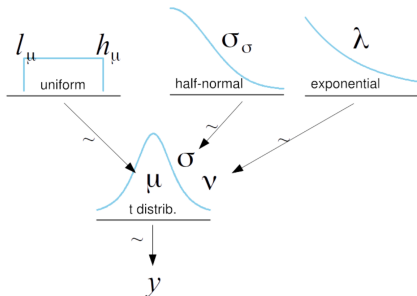
- **Student's t-distribution** has 3 params:
  1. Mean  $\mu$ 
    - Doesn't always exist
  2. Scale  $\sigma$ 
    - Similar to std dev, but doesn't always exist
  3. Degrees of freedom  $\nu \in [0, \infty]$ 
    - “Normality parameter”: controls distribution normality
    - $\nu = 1$ : heavy tails, no mean (Cauchy)
    - $\nu \rightarrow \infty$ : Gaussian
- Student's t has **heavy tails** (high kurtosis)
  - Values more likely far from mean compared to Normal



# Chemical Shift: Use Student's t-dist (1/3)

- Use Student's t-distribution instead of Normal in model for chemical shift

$$\begin{cases} \mu \sim U(l, h) \\ \sigma \sim \text{HalfNormal}(0, \sigma) \\ \nu \sim \text{Exp}(\lambda) \\ y \sim \text{StudentT}(\mu, \sigma, \nu) \end{cases}$$



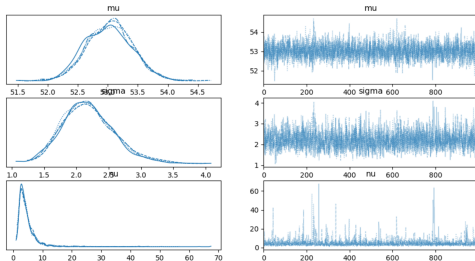
# Chemical Shift: Use Student's t-dist (2/3)

- Decrease  $\nu$  (**less Gaussian**) instead of increasing mean and standard deviation
  - $\mu$  similar to Gaussian estimate
  - $\sigma$  smaller
  - $\nu \approx 5$  (not very Gaussian)
- **Estimation more robust**
  - Outliers have less effect

```
[102]: # Use a Student-T model.  
with pm.Model() as model_t:  
    mu = pm.Uniform("mu", 40, 75)  
    sigma = pm.HalfNormal("sigma", sigma=10)  
    # A student with nu = 30 is close to a Gaussian.  
    nu = pm.Exponential("nu", 1/30)  
    #  
    y = pm.StudentT("y", mu=mu, sigma=sigma, nu=nu, observed=data)  
    idata_t = pm.sample(1_000)
```

Auto-assigning NUTS sampler... \*\*\*

```
az.plot_trace(idata_t);***
```



```
[104]: az.summary(idata_t, kind="stats").round(2)
```

```
[104]:
```

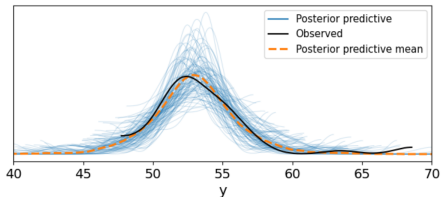
|       | mean  | sd   | hdi_3% | hdi_97% |
|-------|-------|------|--------|---------|
| mu    | 53.03 | 0.38 | 52.35  | 53.76   |
| sigma | 2.19  | 0.40 | 1.45   | 2.95    |
| nu    | 4.65  | 3.92 | 1.20   | 9.20    |



# Chemical Shift: Use Student's t-dist (3/3)

```
[105]: # Compute 100 posterior predictive samples.  
y_ppc_t = pm.sample_posterior_predictive(idata_t, model_t);  
Sampling: [y]  
Sampling ... 100% 0:00:00 / 0:00:00
```

```
[106]: ax = az.plot_ppc(y_ppc_t, num_pp_samples=100, mean=True)  
ax.set_xlim(40, 70);
```



- PPC fits better than Normal model
- Plot is “hairy” because KDE is estimated only in data interval and 0 outside

# Tutorial

---

- [Robust modeling](#)

- Posterior-Based Decisions
  - Chemical Shift: Example
  - Posterior Predictive Checks
  - *Groups Comparison*

# Group Comparison

---

- **Group comparison** tests for statistically significant results between *treatment* and *control group*
- E.g.,
  - *How well do patients respond to a new drug vs a placebo?*
  - *Is there a reduction in car accidents after new traffic regulation?*
  - *Does college student performance improve without cellphones at school?*
- **Effect size** quantifies the difference between two groups
  - From “*does it work?*” (hypothesis testing) to “*how well does it work?*” (estimate effect size)

# Bogus Control Groups

---

- **Always ask for the baseline** used for comparison, when something is claimed to be harder/better/faster/stronger
  - E.g., sell sugary yogurts to boost the immune system by comparing it to milk
    - \* Better control group: less sugary yogurt
- **Placebo** is a psychological phenomenon where a patient experiences improvements after receiving an inactive treatment
  - Using a placebo is better than “no treatment”
  - Shows difficulty in accounting for all factors in an experiment

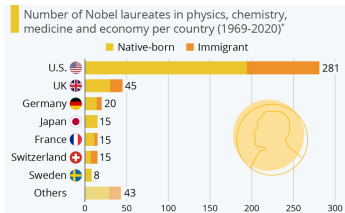
# Group Comparison Bayesian-Style

---

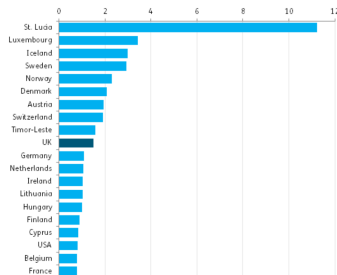
- **Frequentist approach**
  - Compare p-value of difference of means in each group
- **Bayesian approach**
  - Compare posterior distribution of means between groups using:
    - \* Plot of posterior
    - \* Cohen's d
    - \* Probability of superiority

# Sample Size Effect

- **Sample size effect** is the impact of the number of observations on statistical results (e.g., p-values, confidence intervals)
- **Small sample effect**
  - Large mean difference might not be statistically significant (low p-value)
  - Estimates (e.g., means, proportions, correlations) fluctuate widely due to high sampling variability
  - Outliers have a disproportionate influence
  - Inference is unstable: results may not replicate
- **Large sample:**
  - Tiny mean difference can be highly significant (small p-value) but meaningless
- E.g., Cohen's d, probability of superiority



Nobel Laureates per million 2016 population by country of birth (top 20 countries)

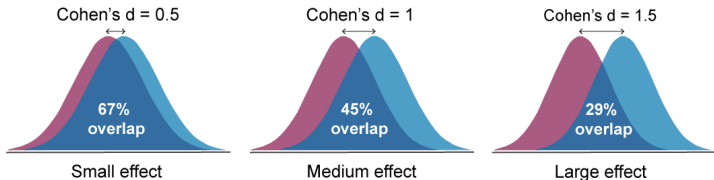


# Cohen's d

- **Cohen's d** is the difference of means relative to pooled standard deviation

$$\frac{\mu_2 - \mu_1}{\sqrt{(\sigma_1^2 + \sigma_2^2)/2}}$$

- Normalizes effect by variability for pooled std dev
- Variability of each group normalizes mean difference
- Similar to a Z-score, number of std dev values differ



- **Bayesian approach**
  - Compute posterior distribution of means and std → formula
  - Compute distribution of Cohen's d → summary statistics